

Performance analysis of hybrid off-grid renewable energy systems for sustainable rural electrification

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ABSTRACT

Delivering sustainable power to rural communities in sub-Saharan Africa, particularly in Somalia, where many areas are far from the electrical grid and suffer from energy shortages, necessitates the deployment of appropriate technologies. This study evaluates the techno-economic and environmental viability of a hybrid renewable energy system (HRES) comprising a 15 kWp photovoltaic (PV) generator, 10 kW wind turbine (WT), 25 kW diesel generator (DG), and a 72-kWh battery energy storage system (BESS). The HRES is proposed to supply sustainable power to a rural community in Somalia. The techno-economic assessment of the proposed HRES and a comparative analysis of other configurations—namely the PV/WT/BESS and PV/DG/BESS systems is conducted using HOMER and MATLAB software. The results indicate that the proposed PV/WT/DG/BESS HRES demonstrates superior techno-economic performance compared to the other configurations. The lowest net present cost of \$96,899.16 and the levelized cost of energy (LCOE) of \$0.090/kWh were achieved. This system attains 25 % excess electricity, maintains a 0 % unmet electric load, and demonstrates robust environmental performance with a renewable penetration rate of 91.8 %. Approximately 53.34 % of greenhouse gas emissions are reduced compared to the PV/DG/BESS configuration. The study confirmed the significant impacts of all sensitivity parameters on operational costs, LCOE, fuel prices, fuel intake, and renewable fraction. The overall results indicate superior performance in the optimal system case, offering a feasible and suitable choice for the sustainable electrification of rural communities.

1. Introduction

Due to rapid population growth, urbanization, economic development, and technological advancements, coupled with the imperative of environmental protection, global energy demand has notably increased [1–3], necessitating the development of new power generation capacities [4]. The global population is predicted to grow by almost one-fourth from 7.8 billion in 2021 to 9.7 billion in 2050, requiring energy services for more people. However, the annual population growth is projected to decrease from approximately 1 % in 2020 to 0.5 % in 2050

[5]. The populations of Asia and sub-Saharan Africa (SSA) are predicted to double from about 13 % by 2022 to 26 % by 2050 [5]. Lack of access to affordable, reliable, and sustainable energy sources remains a significant challenge in many developing countries, particularly in SSA and South Asia [6]. According to global energy data, over 675 million people lack access to electricity infrastructure. Over 80 % of them reside in rural areas, particularly in SSA countries. Researchers project that around 660 million people will still lack access to electricity by 2030, with approximately 85 % of them residing in SSA [7]. The global energy demand is projected to increase by 1.3 % annually between 2020 and

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2030 if no changes are made to energy policies and efficiency [8]. In line with the International Energy Agency (IEA) stated policies scenario, energy-related carbon dioxide (CO₂) emissions exhibited 1.9 % growth in 2018 and are expected to reach around 36 gigatons by 2050 [9]. To limit temperature rise below 2 °C or preferably 1.5 °C as per the Paris Agreement, reducing energy-related emissions to zero by 2050 is required, which demands a fundamental shift to renewable energy resources (RERs), energy efficiency, and carbon capture and storage technologies. Achieving this transition requires significant effort from all stakeholders, including policymakers, investors, and individuals [5]. However, the extensive use of fossil fuels to meet current energy demands has created an urgent need for alternative sources, prompting researchers and policymakers to seek sustainable and long-term solutions. Fossil fuel consumption also contributes to global warming [10–12]. To address this global energy problem, RERs are projected to play a crucial role in providing green energy [13,14]. Thus, the world prioritizes RERs to reduce CO₂ emissions, improve energy access, support sustainable development in remote areas, create job opportunities, facilitate infrastructure development, and enhance local health conditions [15,16]. The cost of renewable power generation has significantly decreased over the past decade, as depicted in Fig. 1, due to technological advancements, supply chain competitiveness, and supportive government policies [17]. The assessment showed that energy efficiency and RE technologies are critical to the transition, and their interconnections should be considered. The shift towards RE is supported by various factors, such as beneficial economics, abundant resources, easily scalable technology, and significant socioeconomic benefits. Adopting RE can fulfill two-thirds of the world's energy demand, significantly reducing greenhouse gas (GHG) emissions [13,18–20]. Thus, prioritizing RERs is crucial for ensuring energy security and fostering sustainable development globally.

1.1. Literature review and background

Researchers have recently conducted comprehensive studies on hybrid renewable energy systems (HRES) in the context of sustainable rural electrification [6,21,22]. These systems show functionality when connected to a grid and operating off-grid. Numerous models have been developed, employing various techniques, including classical methods, Hybrid Optimization of Multiple Energy Resources (HOMER), MATLAB, genetic algorithms, and particle swarm optimization. These models aim to enhance cost-effectiveness and address the specific requirements of underdeveloped rural communities [23–25]. In this context, various techno-economic analyses have been conducted. A grid model with solar and wind power, along with utility-scale batteries as backups, is simulated by the HOMER Microgrid Analysis Tool. The LCOE and societal cost of electricity (SCOE) were estimated at different locations through 495 trials, considering carbon taxes, emission savings, and renewable

fractions (RF). The optimal RF for each site was determined, with ranges of 34 %–48 % for New York, 0 %–34 % for San Diego, 55 %–75 % for Milwaukee, and 21 %–65 % for Texas [26]. A techno-economic analysis of meeting the energy requirements of off-grid vacation homes with hybrid energy systems (HESs) photovoltaic/wind/fuel cell (PV/WT/fuel cell) considering the beneficial solar and wind energy potential of Çeşme, İzmir, a location with seasonal vacation residences, 24 simulations were conducted using HOMER software [27]. The integration of HRES is dependent on government support and community engagement. Diesel is identified as the most expensive option in LCOE comparisons (\$0.92–1.30/kWh), followed by hybrid solar/diesel (\$0.54–0.77/kWh) and solar PV (\$0.40–0.61/kWh). Costs can be reduced, and policy gaps can be addressed through government incentives, particularly in developing countries, where HRES is considered a faster and more cost-effective alternative to grid expansion [6]. The proposed model is optimized to select various wind turbine technologies. Case study results show a 5 % reduction in planning costs, depending on wind speed, and a 700 kWh decrease in diesel generator (DG) energy production, reflecting increased renewable integration and lower emissions [28]. The proposed HRES, consisting of PV, fuel cells, and a biogas generator, addresses Pakistan's power shortage by offering a cost-effective solution for rural areas. Robustness, sensitivity, and breakeven analyses confirm the system's feasibility and reliability in delivering electricity. The research shows a cost of energy of \$0.41/kWh and a total net present cost of \$199,081 [29]. The promise of addressing the energy transition has been demonstrated through the utilization of HRES, with simulations being conducted employing various techno-economic modeling tools. Among these tools, the HOMER software is the most widely employed [30,31].

Numerous studies have examined HRES in different locations globally. The power plants in Sabah, Malaysia, were evaluated, creating an optimal RE system for a remote village. Evaluation of on-grid and off-grid scenarios proved that the design met load demands, ensured continuous operation, and reduced emissions. The study highlights the importance of considering comprehensive parameters and the significance of hybrid system implementation [25]. The PV/WT hybrid system generated efficient and cost-effective hydrogen storage under various climatic conditions. The system utilizes the abundant wind and solar energy resources of five Egyptian cities with fixed wind turbines and PV arrays while adapting to specific climatic conditions based on location [32]. In a study led by G. Veilleux *et al.*, Koh Jik Island's rural microgrid in Thailand was upgraded using costly Li-ion batteries, achieving a 6.3-year payback period and a 15.3 % internal rate of return. Emphasizing storage size, reduced generator runtime, and lower battery prices enhance affordability and eco-friendliness. Investors valuing experience or accepting lower indicators are drawn to such projects, and the lessons from Koh Jik are shaping electrification strategies in neighboring Asian countries [33]. Researchers, led by O.D.T. Odou *et al.*, assessed HRES

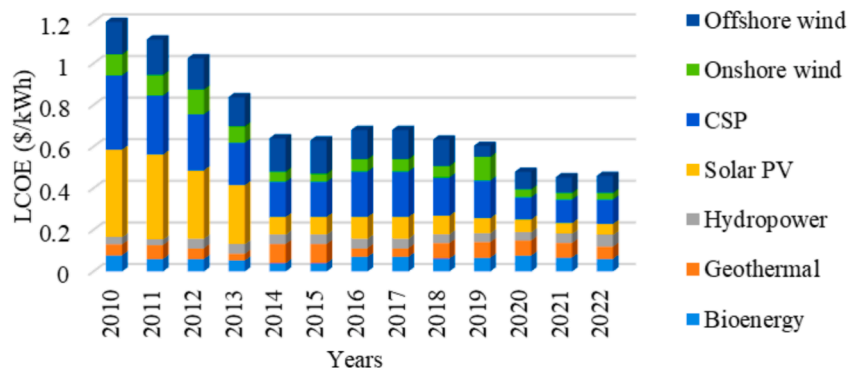


Fig. 1. Variations in renewable power generation costs [17].

potential for rural electrification in Fouay village, Benin, using HOMER software. Results favored a hybrid solution of PV and DG with a battery storage system [30]. In Ref. [34], the diesel fuel consumption, energy expenses, and yearly CO₂ emissions were systematically examined in a DG-only system. A parametric analysis was conducted to assess the influence of PV/WT/DG/battery capacities on both CO₂ emissions and the LCOE across diverse locations in Turkey. According to Ref. [35], the research examines six configurations to determine the best solution based on criteria specific to the Afghanistan community. The most effective configuration is characterized by a capital cost of \$5962, a net present cost (NPC) of \$10108, total O&M costs of \$320.69, and an LCOE of \$0.190 per kWh. Ref. [36] assessed the performance of PV/WT/DG/battery systems and converters intended for a large resort in the South China Sea, Malaysia. These investigations focused on evaluating techno-economic aspects using HOMER software. Ref. [37] reports a substantial 62 % reduction in CO₂ emissions compared to kerosene and 67 % compared to the grid-connected system in northern Bangladesh through implementing PV/WT/DG with battery storage, as assessed and optimized using HOMER software.

The current study on HRES affirms their potential to address off-grid instability and reduce emissions. However, several critical research gaps exist, including the lack of understanding regarding location-specific feasibility, methodological considerations in evaluating HRES, and insufficient attention to the environmental implications of PV systems within HRES configurations. Addressing these research gaps is essential for advancing the implementation and optimization of HRES, ensuring their effectiveness, sustainability, and environmental compatibility across different geographic locations.

1.2. Energy status, research gap and contributions

In Somalia, access to electricity remains a significant challenge, with only 49 % of the population having access, according to World Bank data. The country aims to increase its electricity capacity to 1,043 MW between 2022 and 2027, with the electrification rate expected to rise to 75 % [4]. As the private sector continues to invest in various sectors, including buildings, irrigation, cold storage, and small industries, the demand for energy is projected to rise, leading to expansion. A key issue is power sustainability, particularly in remote rural areas where access is even more limited. The country's energy landscape heavily depends on imported oil, which is used for transportation and electricity generation. Additionally, firewood remains a primary source of cooking fuel. The current electricity generation mix is dominated by fossil fuels, which account for 87.8 % of the total, while RE sources contribute only 12.2 % [4]. This heavy reliance on imported oil poses economic challenges, as it depletes the country's hard currency reserves. The widespread use of firewood and charcoal for cooking has led to significant deforestation, exacerbating environmental concerns. The lack of a national grid infrastructure means that most regions rely on diesel generators for power supply. These generators are costly to operate and maintain and contribute to high levels of GHG emissions, impacting the economy and the environment. Efforts to improve the energy sector in Somalia must address these comprehensive challenges. Enhancing the RE share, improving energy efficiency, and developing a comprehensive national grid infrastructure are crucial steps towards achieving sustainable and reliable power access across the country.

The literature review confirms that the combination of RERs, diesel generators, and battery storage effectively mitigates off-grid instability in both urban and rural communities. However, the feasibility of implementing HRES relies on location-specific RE potential and meteorological conditions. Therefore, it is crucial to investigate various sites to evaluate system viability, with the selected methodology playing a significant role in obtaining accurate findings. In addition, the practical application of HRES, particularly when integrated with multiple RERs, is currently extensive. Contrary to the options outlined, it is essential to bear in mind that most researchers have mostly overlooked the

environmental implications of PV systems. According to an in-depth review carried out by the authors, it is evident that no study has been conducted on HRES in Somalia. However, a knowledge gap exists concerning the performance analysis and optimal design of PV/WT/DG systems with a battery energy storage system (BESS) hybrid configuration in Somalia [38]. This study aims to fill this knowledge gap, which has hampered Somalia's economic and sustainable development. Without consensus regarding the optimal sizing and feasibility study of sustainable electrification in Somalia's communities, a potential solution remains uncertain. As grid extension is impractical in this area, integrating RESs with an existing DG is explored as the optimal solution. We propose a system model introducing a PV/WT/DG/BESS configuration incorporating a bidirectional inverter. This study aims to provide a comprehensive analysis focusing on the optimal sizing and performance evaluation of HRES for sustainable rural electrification in Somalia. The contribution of this study has been summarized as follows:

- Developing a model for hybrid energy systems entails exploring different combinations of RE sources, including PV, WT, DG, and BESS, aimed at fulfilling community energy demand.
- To comprehensively analyze various system configurations, this study will compare three cases for each configuration, evaluating them from the perspectives of reliability, technical performance, sustainability, and economic viability.
- Conducted a comprehensive analysis to optimize electricity costs and generation capacity to implement HRES in sustainable rural electrification. This analysis focused on achieving optimal performance, sensitivity to varying conditions, and minimizing environmental impact.
- This study examines the environmental impact of PV systems to understand the relationship between PV power generation and its associated emissions.

The rest of this paper is organized as follows: The proposed integrated modeling of HRES and its methodology are presented in Section 2. Results and related discussions are provided in Section 3. Finally, concluding remarks are provided in Section 4.

2. Methodology

This section provides a detailed rationale for the multi-objective optimization process and introduces the study's tool. Technical and economic data, design specifications, and mathematical models for microgrid RE systems are discussed. The process involves selecting the optimal location, analyzing the energy demand, and optimizing the system. This study focuses on integrated microgrid HRES, including PV panels, wind turbines, batteries, converters, and DG. PV panels and wind turbines meet the community's energy requirements, whereas DG remains on standby. Surplus power is stored in batteries, and the DG is used to satisfy the load when the generated power is insufficient, such as at night or on cloudy days. The batteries and WT compensate for any shortages by supplying stored electricity.

Fig. 2 depicts the comprehensive methodology for optimizing HRES using HOMER Pro® software. To ensure system viability, a proprietary derivative-free algorithm ranks system configurations based on economic criteria to assess the feasibility of designs within the search space and sensitivity variables. The results are presented in an optimization table comparing key metrics such as NPC, LCOE, excess electricity, and power generation.

2.1. Description and load data of the site under study

The present study focuses on a Dismareb district in Somalia, which serves as the administrative center of Galmudug State's official hub. The city lies 511 km from Mogadishu, the capital of Somalia, and is strategically positioned on a highway that connects the country's northern

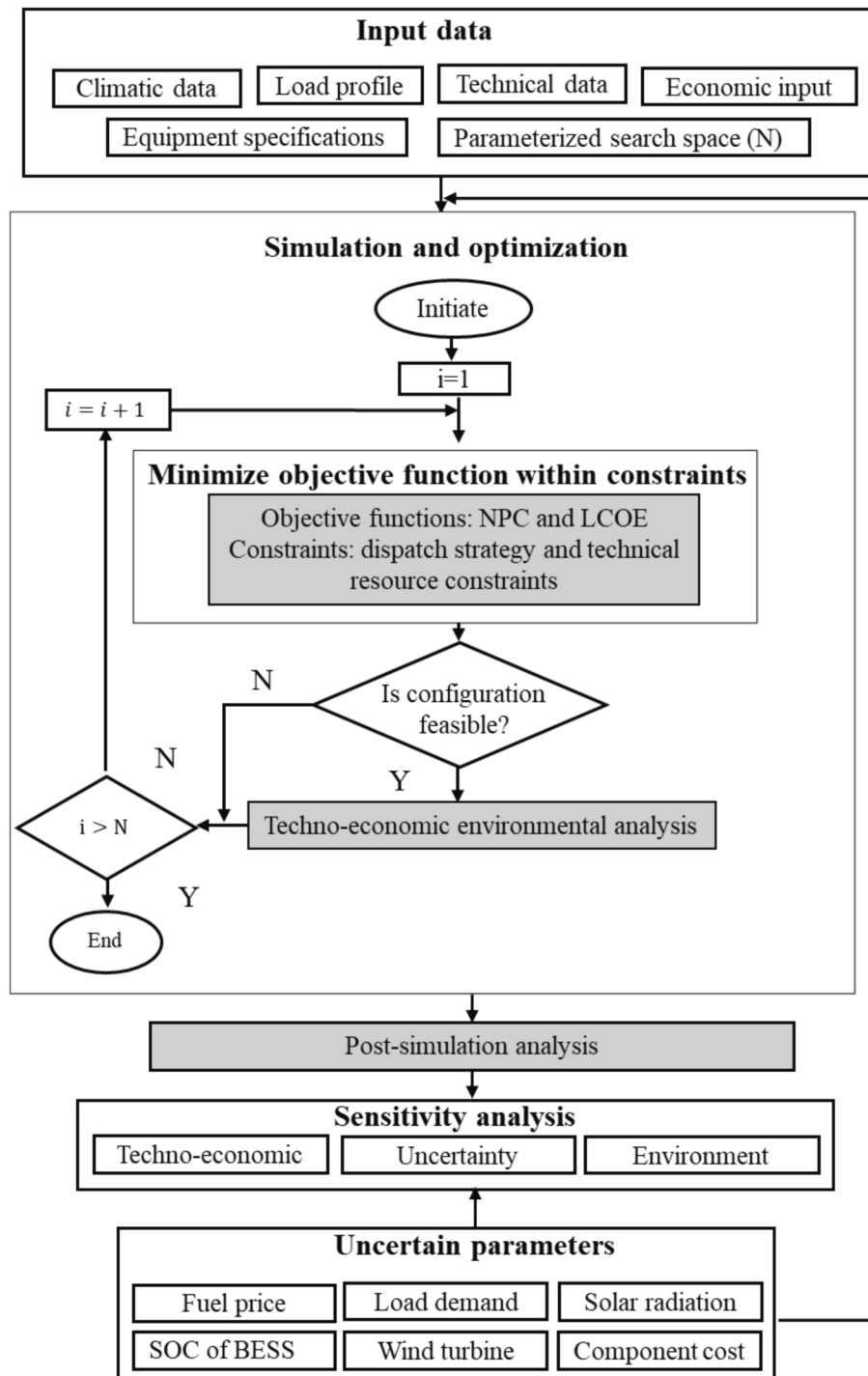


Fig. 2. The methodology adopted in the analysis of the multi-energy community HRES.

and southern regions. A thorough community analysis determined the typical electrical loads and their characteristics. The results of this assessment were used to create a load profile, as detailed in Table 1. The analysis of 82 off-grid households is undertaken, and the collected data is utilized to calculate community demand. The daily energy consumption reaches 136.12 kWh per day, as determined by the household appliances in each property. The peak load, demonstrating 22.01 kW, is determined using HOMER software to ensure the system's dependability in meeting daily load demands. To accurately estimate the peak load of the proposed configuration, 13.09 % random time-step and 16.832 % random day-to-day variations were entered into HOMER to account for

Table 1

Electric load computations for a typical domestic village in summer.

Load	Power (watts)	Number in use	Operating hours	Total load (Wh)
LED lights	10	4	6	240
Fans	40	3	7	840
Refrigerator	15	1	24	360
Washing machine	220	1	1	220
The total load for a single residential unit				1.660 kWh
The cumulative load of 82 residential units				136.12 kWh

the fluctuating daily power demand of the community load. The following analysis revealed baseline and scale values of 272.36 kWh and 136.38 kWh, respectively. Additionally, observations indicated that the system's baseline electric load was 39.54 kW, with a scale load of 20.01 kW. HOMER Pro® primarily generated the system's base load, as illustrated in Table 2.

Analysis of the daily load profiles indicated the presence of distinct demand peaks during the morning and evening periods, with a particular emphasis on residential households. Daytime demand displayed a reduced scale owing to the probability of most residents being occupied with their professional duties. However, the overnight load was increased when all residents were present at their respective residences. Fig. 3 shows the selected community load profiles, which consisted of hourly and monthly load data from the power load demand. In April and May, the mean daily power consumption was recorded at 85.5 kWh, while the average monthly power consumption was 2,565 kWh. Energy usage in rural residential areas was found to be lower than that in urban residential districts. The primary contributors to the total power use of the community were residential lighting, fans, and freezers. In the community, there was a significant use of power due to the prevalent usage of lighting, fans, and freezers as essential everyday utilities. The electric load was accurately assessed by considering the power usage of the residences during the peak summer season. The present study subsequently assessed the electrical demand and performed a techno-economic assessment of an HES considering the load demand and environmental considerations.

2.2. Meteorological data assessment

Somalia receives an average annual sunshine duration of 2,900 to 3,100 h, equivalent to approximately 8 to 8.5 h of sunlight per day. The region's ambient temperature, which averages 27 °C year-round, provides favorable conditions for the longevity and efficiency of PV systems. In addition, Somalia's annual average solar radiation ranges from 5.62 to 6.67 kWh/m²/day. Based on this data, the average annual solar energy potential is approximately 6.26 kWh/m²/day, positioning the country as one of the sunniest regions globally [4]. These factors indicate Somalia's significant potential for solar energy generation, given its proximity to the equator, where sunlight is abundant throughout the year. The meteorological data used in this study were gathered from the NASA Langley Research Center website by defining the geographical coordinates (5° 32' 0" N latitude and 46° 32' 0" E longitude) of Dumsareb. The data presented in Fig. 4 depict the monthly variations in average sun irradiation and the accompanying clearness index. The amount of solar radiation reaching the Earth's surface is influenced by overcast circumstances and sky clarity, which are highly unpredictable. The mean yearly solar irradiation was 6.31 kWh/m²/day. This finding implies a massive potential for the adoption of solar PV in this region, regardless of the present underutilization of these resources. The monthly average air temperatures in Dumsareb revealed variations, with the highest and lowest values recorded in April (29.29 °C) and January (26.15 °C), respectively, resulting in a monthly average of 27.90 °C (refer to Fig. 5).

The proposed system comprises four main components: a wind turbine, PV array, diesel generator, and converter. The wind speed varied from 10.79 m/s (July) to 10.43 m/s (August), with 7.77 m/s (annual average), as shown in Fig. 6. Variations in wind speed patterns are typically attributed to several factors, including geographical

characteristics and vegetation coverage. The HOMER analysis tool generates a yearly synthetic per-hour dataset for each year based on the entered mean wind speed value for each month. This dataset incorporates cutting-edge parameters such as the 1-hour value, autocorrelation factor of 0.85, diurnal pattern strength intensity of 0.25, peak wind speed occurrence at 15 h, and Weibull shape parameter of 2. Table 3 presents comprehensive details regarding the cost components necessary for the simulation. This study utilized HOMER software to perform a techno-economic feasibility evaluation on various system models involving the proposed configuration. The objective is to determine the optimal combination of HES for supplying power to the designated community facility.

2.3. Mathematical modeling and system specifications for HRES components

Fig. 7 depicts the proposed schematic design, in which the electrical load requirements are connected to an alternating current (AC) bus. The system incorporates three resource components: PV, WT, and DG, all integrated with a battery via a bidirectional converter. The PV and Li-ion batteries were connected to a (DC) bus connected to the AC supply through a bidirectional converter. The primary objective of this system is to provide electricity through a PV array and a wind turbine. Any surplus energy beyond the electrical load demand is stored in the Li-ion battery. It is important to note that this study only considered residential AC loads. The proposed HRES comprises a PV array, wind turbine, DG, and Li-ion battery storage system. The technical and economic details are outlined in the following sections.

2.3.1. Solar PV model

A PV array generates electrical energy to meet the diverse power demands of clinical, residential, and commercial loads. Solar irradiation levels, cell temperature, and the location's geographical characteristics influence a PV system's power capacity. The optimal year-round tilt angle is calculated to be 7.6° from the horizontal. However, to facilitate the natural cleaning of solar panels by rain, installers typically set the tilt angle to a minimum of 10°. This study used a flat-plate residential PV array with a nominal capacity of 1 kWp under standard testing conditions (STC) and a panel efficiency of 20.4 %. Additionally, the power temperature coefficient for this module is found to be −0.5 %/°C, whereas the working temperature is 47 °C. The derating factor for the panel was determined to be 89 %, and the ground reflectance, which represents the proportion of reflected solar irradiation received by Earth's solid surface, was taken to be 20 %. The PV array had a lifespan of 25 years.

Given that the projected lifespan of the project is 25 years, there is no need to allocate funds to the replacement costs of the PV array. The component cost parameters are listed in Table 3. The PV power output was determined using Eq. (1) [2]:

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha_p (T_C - T_{C,STC})] \quad (1)$$

where Y_{PV} refers to the PV power output at STC (kW), α_p denotes the temperature coefficient of power (%/°C), f_{PV} denotes the PV derating factor (%), G_T represents the incident irradiance at STC (1 kW/m²), G_T represents the solar irradiance incident on the PV system (kW/m²), T_C represents the PV cell temperature (°C), whereas $T_{C,STC}$ represents the PV cell temperature at STC (25 °C).

2.3.2. Wind turbine model

Generating electrical energy from wind involves converting wind power into mechanical power using a WT, which is then transformed into electrical energy. The height of the WT significantly impacts the electricity generated, affecting the overall output. For the current analysis, a WT with a rated output of 1 kW and a hub height of 20 m was

Table 2

Summary of the scale and baseline electric load metrics.

Metric	Baseline	Scaled
Average (kWh/day)	272.36	136.38
Average (kW)	11.35	5.68
Peak (kW)	39.96	20.01
Load factor	0.28	0.28

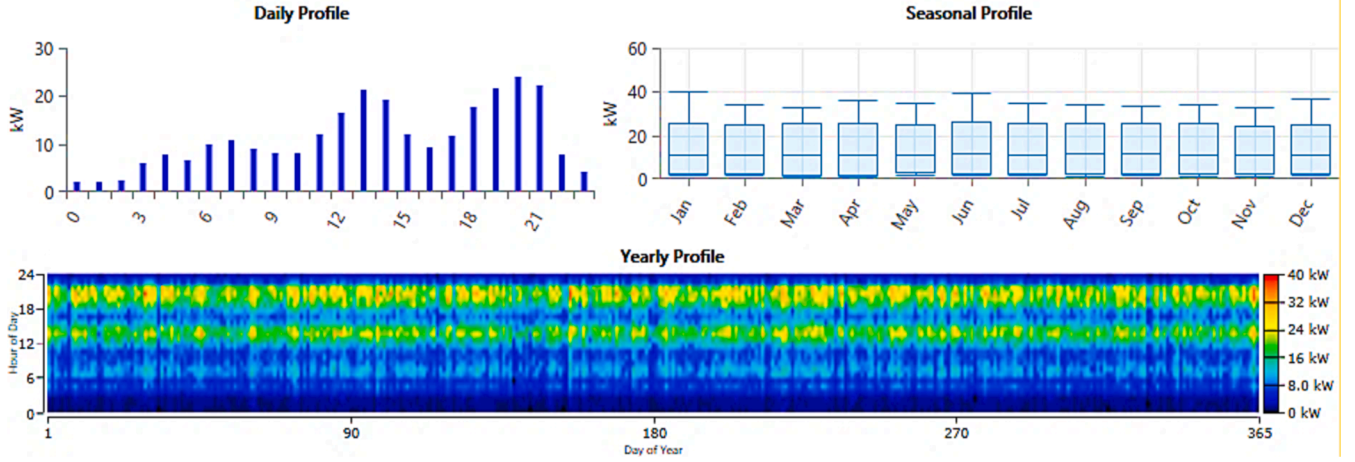


Fig. 3. Daily and seasonal load demand.

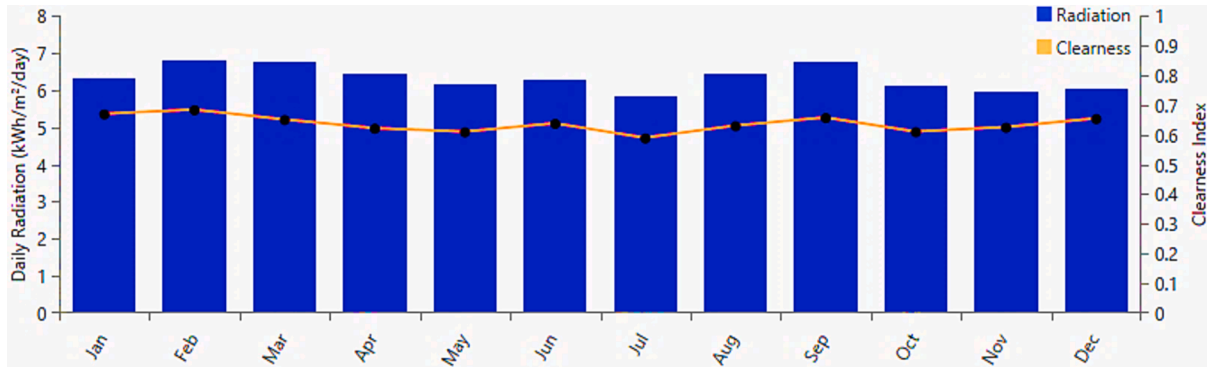


Fig. 4. Monthly average global horizontal radiation and clearness index for Dumsareb.

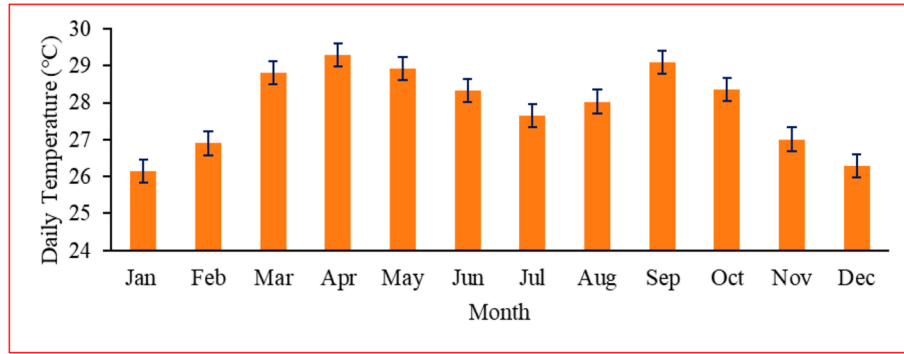


Fig. 5. Monthly variations in ambient temperature in Dumsareb.

selected. Table 3 presents the cost parameter components associated with the WT over its 20-year lifetime. The WT output in the study area can be computed using Eq. (2) [2]:

$$P_{WT} = \begin{cases} 0, & 0 < V \leq V_{C,in} \text{ and } V \geq V_{C,out} \\ \frac{P_r(V - V_{C,in})}{(V_r - V_{C,in})}, & V_{C,in} \leq V < V_r \\ P_r, & V_r \leq V < V_{C,out} \end{cases} \quad (2)$$

where P_r , V , $V_{C,in}$, V_C , and V_{OUT} represent the rated output power of the WT, wind speed, cut-in wind speed, and cut-out wind speed, respectively.

The mechanical power of the WT can be calculated using Eq. (3).

$$P_m = \frac{1}{2} \times \rho \times A \times V^3 \quad (3)$$

where ρ is the density of the air (1.22 kg/m^3), A is the area moved by the rotor (m^2), and V is the wind speed (m/s). The electrical power of the WT can be determined in Eq. (4), where C_p refers to the power coefficient of a WT.

$$P_e = \frac{1}{2} \times \rho \times C_p \times A \times V^3 \times 10^{-3} \quad (4)$$

2.3.3. Battery energy storage system model

The BESS is an energy storage and delivery system that ensures a hybrid system's reliable and efficient operation using RE sources. The

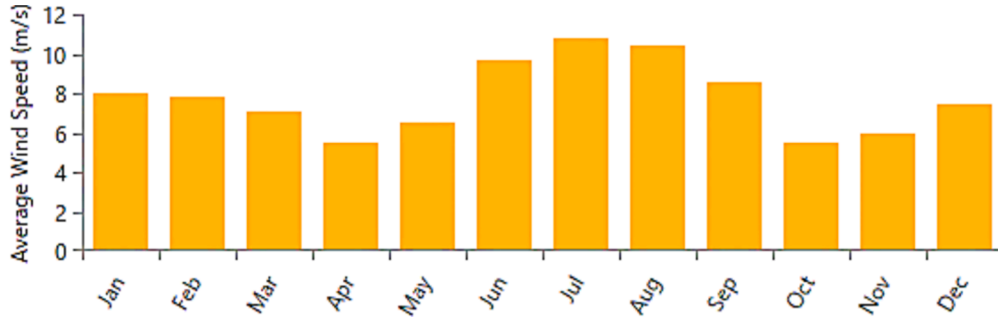


Fig. 6. Temporal evolution of monthly wind speed variations in Dusmareb.

Table 3

Cost parameters and technical specifications of the components.

Components	Power capacity	Capital cost	Replacement cost	O & M cost
PV system	1 kWp	\$480/kW	\$480/kW	\$5/kW/year
Wind turbine	10 kW	\$8,000/kW	\$8,000/kW	\$500/kW
Diesel generator	1 kW	\$500/kW	\$500/kW	\$0.030/kW/hr.
Battery	1 kWh	\$280/kWh	\$280/kWh	\$5/year
Converter	1 kW	\$200/kW	\$200/kW	\$1.5/year

primary purpose of this analysis is to store surplus power generated by the system and supply energy to the load when generation is insufficient. The BESS selected for this analysis had a rated nominal voltage and capacity of 6 V and 1 kWh, respectively. It maintains a minimum state of charge of 20 % and has an energy throughput of 280,970 kWh. Additionally, it exhibited 90 % round-trip efficiency. Table 3 presents the cost variables associated with this component. The string size of the BESS was projected to consist of 35 parallel batteries. Each battery has a designated expected life, known as the throughput of 280,970 kWh. In addition, these batteries are accompanied by the stated cost parameters. The mathematical representation of the battery is given in Eq. (5) [2]:

$$C_{wh} = \frac{E_L \times AD}{\eta_{inv} \times \eta_b \times DoD} \quad (5)$$

where E_L represents the total energy demand in kW. AD represents the autonomy days. η_{inv} represents the inverter efficiency, η_b represents battery efficiency, which is considered to be 90 % round-trip efficiency,

and DoD represents battery depth of discharge.

2.3.4. Power converter model

The power converter operates in two modes: an inverter converting DC to AC, or a rectifier converting AC to DC. It ensures a continuous electric current between the AC and DC bus components. The choice of converter typically depends on the power output of the solar PV array. Table 3 lists the cost parameters of the converters. The inverter input is required to have a 20-year lifetime and an efficiency of 95, whereas the rectifier input has a relative capacity of 100 % and an efficiency of 95 %. The power rating of the converter is determined using the following Eq. (6).

$$P_{con} = \frac{P_{peak}}{\eta_{con}} \quad (6)$$

where η_{con} represents the efficiency of the converter, and P_{peak} denotes the peak load at the point of consumption.

2.3.5. Diesel generator model

DG, a provider of electrical power in HESs, emphasizes fuel efficiency. Often deployed as backups alongside RESs, it counters the variable nature of RESs. Given RESs' reliance on environmental factors, a DG with a minimum 50 % load ratio is recommended. DGs have a 15,000-hour operational lifespan. Their cost details are in Table 3. HOMER uses fuel curves and linear correlations to calculate fuel intake, and fuel consumption for power generation can be determined using Eq. (7) [38]:

$$F = (F_{0,DG} \times Y_{DG}) + (F_{1,DG} \times P_{DG}) \quad (7)$$

where F denotes the DG fuel intake (units/h) at the current time step, F_0 ,

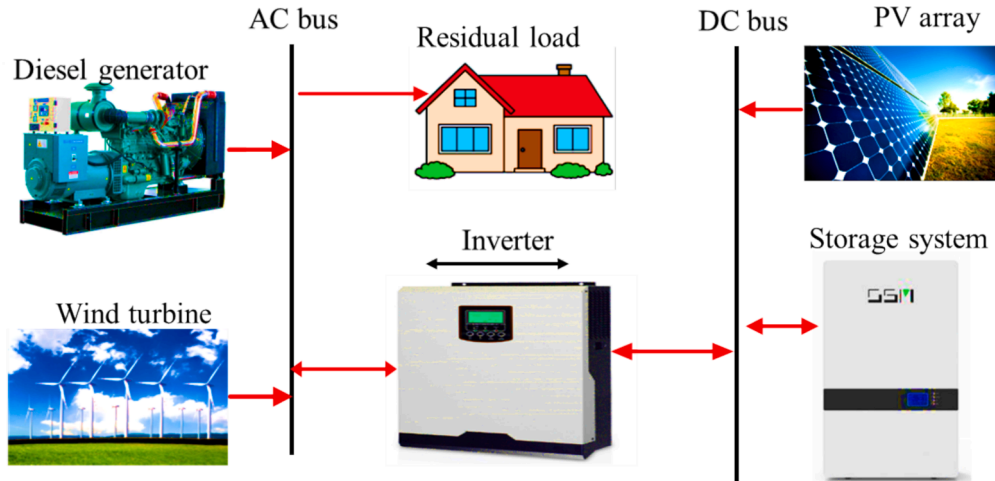


Fig. 7. Schematic representation of the proposed HRES.

DG is the coefficient of the generator fuel curve intercept (units/kW) (taken as 1.40 L/hr rated), Y_{DG} is the rated capacity of the generator (kW), $F_{1,DG}$ is the fuel curve slope for the generator (units/kW) (0.236 L/h/kWh), and P_{DG} is the power generator output (kW).

2.3.6. Economic determinants

In this study, HOMER was utilized due to its techno-economic effectiveness, making it a widely used tool for simulating hybrid energy sources globally. HOMER has been extensively applied in planning, conducting feasibility tests, scheduling, and analyzing system inputs in both off-grid and grid-connected systems to determine the optimal solution for sustainable electrification. The proposed configurations are ranked based on the NPC and LCOE, which serve as crucial economic parameters. These parameters guide the assessment of different system models to identify the most efficient solution. The study aims to minimize the NPC and LCOE, which are computed using Eq. (8) [39]:

$$NPC = \sum_{i=1}^N \frac{NC_i}{(1+R)^i} - TInvC \quad (8)$$

where NC_i denotes the net cash flow in year i , R signifies the discount rate (%), and $TInvC$ represents the total investment cost. A project is considered financially feasible when the NPC is more than zero. Calculating the capital recovery factor (CRF) is essential in financial analysis and project evaluation. CRF is determined as Eq. (9):

$$CRF(i, t) = \frac{i(1+i)^t}{(1+i)^t - 1} \quad (9)$$

In this study, the salvage value (S) has been considered. The S value denotes the component value after it has been used over the project lifespan, which is evaluated as Eq. (10) [38]:

$$S = C_{rep} \times \left(\frac{T_{rem}}{T_{comp}} \right) \quad (10)$$

where C_{rep} is the replacement cost in \$, T_{rem} is the remaining lifetime of the component in years, and T_{comp} is the number of years that the component is expected to last before it fails. The total annualized cost (TAC) and the LCOE are considered. Using Equations (11) and (12), the LCOE (in \$/kWh) and TAC generate the annualized value of the NPC (in \$/year) as follows:

$$TAC = C_{NPC} \times CRF(i, T_p) \quad (11)$$

$$LCOE \left(\frac{\$}{kWh} \right) = \frac{TAC (\$/year)}{E_{anloaderved}} \quad (12)$$

where $E_{anloaderved}$ represent the gross yearly load (kWh/year) served by the system, and t is the expected lifespan of the project.

In the present study, the economic assessment was determined using the calculation of the payback period, which can be specified by employing the following mathematical Eq. (13):

$$\text{Payback period} = \frac{I}{R + E} \quad (13)$$

where I, R, E represents investment, return, and expenses, respectively.

2.4. Engineering constraints and dispatch strategy

To achieve a realistic optimal result, ensuring reliable power supply meets load demand, sizing optimization necessitates predefined rules. This study utilizes the renowned HOMER software to ascertain appropriate component sizes within energy systems, aligning with electrical load requirements. The primary components of this consideration include modeling, optimization, simulation, and sensitivity analysis. The wind turbine, photovoltaic module, battery energy storage system,

and diesel generator satisfy the load demand. We aim to maintain an uninterrupted electrical power supply by implementing operational buffers equal to 10 % of the total load to accommodate unexpected spikes in energy demand or decreases in RES yield. Additionally, we established WT and PV yields at 50 % and 25 %, respectively, as a percentage of the overall RE output. This decision to set relatively high-RES values accounts for their inherent variability. Throughout the simulation, a peak unmet energy of 0 % and a maximum annual capacity shortage of 0 % were considered.

In RESs with DG units and storage systems, the dispatch strategy is pivotal in minimizing hardware capacity and environmental emissions. This study adopts a load-following (LF) dispatch strategy as the energy management strategy (EMS). Under this approach, the DG unit satisfies the primary load, allocating any surplus energy to high-priority tasks. The RES handles low-priority objectives like charging storage or deferrable loads. Priority is given to RES and battery storage to meet electricity demand. The DG is only activated as a backup when RES and battery storage fall short of meeting the load requirement. This strategy aims to curtail DG's operational and maintenance costs. Fig. 8 presents the complete methodology for the optimum HRES with an EMS using HOMER software. The solar energy generation ($E_{PV}(t)$) and wind energy generation ($E_{WT}(t)$) are compared to the electric load ($E_{Load}(t)$). The BESS is charged first, given the condition of not fully charged ($BESS_{SOC} < BESS_{SOC, max}$) with the excess electricity (dump load) (i.e., $P_{gen}(t) \geq P_d$). After the battery reaches its maximum state of charge, the remaining energy is referred to as the dump load and is used to satisfy the load demand. If the $P_{RE}(t)$ and BESS are insufficient to supply the load demand and the load exceeds the minimum load ratio selected, the failsafe (DG) supplies the required load demand. However, any electric load demand ($P_{Elec}(t)$) below the minimum load ratio backup (DG) that is not met by the $P_{RE}(t)$ and the battery bank at that moment results in an unmet load ($P_{Elec}(t)$) for 25 years. The optimization tool conducts a simulation of the HES. It determines the optimal sizing of the system to minimize the NPC while ensuring that the load demand is met within the specified technical constraints. The results of this simulation are then considered on the testimony of technical and financial perspectives and environmental pollutants.

2.5. Sensitivity analysis

Sensitivity analysis was carried out to examine how specific parameters could impact the optimal pattern. Parameters included diesel fuel price, solar radiation, wind speed, minimum state of charge of BESS, and community load demand. Baseline values varied higher or lower for all sensitivity parameters except for load demand growth. The baseline rates for around the planet fuel price, solar radiation, wind speed, minimum state of charge of BESS (%), and community load demand were \$0.50/L, 6.31 kWh/m²/day, 7.77 m/s, 20 %, and 22.01 kWh/day, respectively. The sensitivity parameters investigated in this study are listed in Table 4. The expected lifespan of the proposed HES is 25 years. A techno-economic feasibility assessment was carried out in this study to identify the most favorable combination of energy-based technologies in various HRES.

3. Results and discussion

The optimal system configuration is analyzed in this section using the RER data and daily power requirements of the community facility equipment. An investigation was undertaken of both the RE potential and the availability of fuel resources for operating the DG to ascertain the viability of HESs. Simulations were conducted for three configurations to evaluate the impact of component combinations on technical, economic, and environmental factors, with these configurations being detailed in Table 5. A realistic optimal system has been achieved through the performance assessments of various HESs using HOMER simulation mechanisms. As previously outlined, sensitivity analysis has

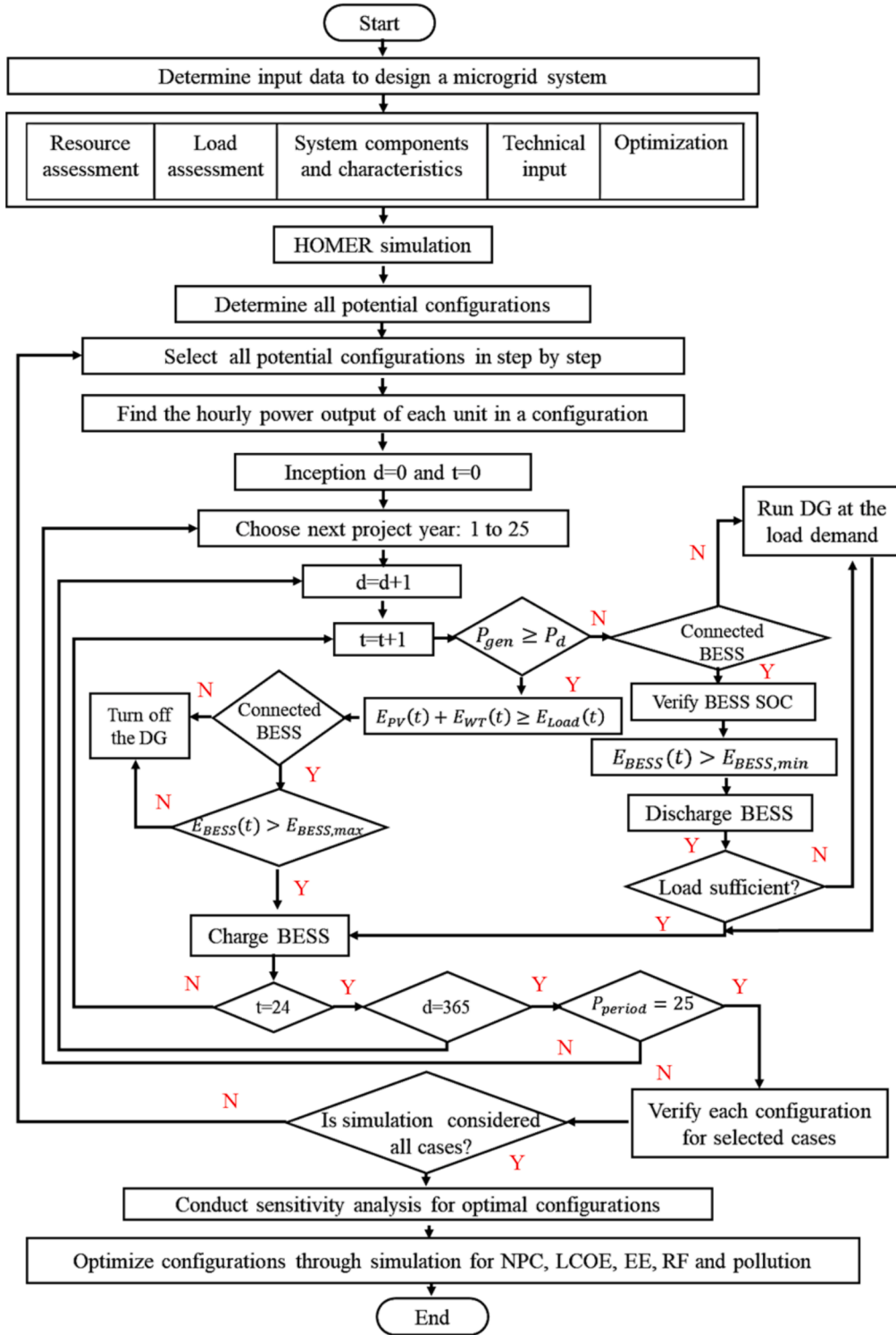


Fig. 8. Workflow representation adopting the LF strategy for optimum HRES.

been conducted using the optimal system and related sensitivity parameters. The optimization results have been presented, and the components, such as the PV, wind, diesel, and BESS, have been sized for optimal performance. All feasible configurations, costs, and performance stages have been considered, and the rankings of systems have been determined based on the overall system performance. The consideration of the capacity of the systems to produce reliable power with minimum environmental impact, as well as evaluation of financial concerns, specifically NPC and LCOE, have been undertaken in assessing

the systems' rankings.

3.1. Performance analysis of the systems

Three configurations, Cases 1, 2, and 3, were used for a techno-economic analysis. This analysis involved system optimization, performance analysis, and a detailed examination of various components. Key aspects evaluated included DG performance metrics such as fuel consumption, operational hours, total power output, PV array WT power

Table 4

Sensitivity parameters used in the analysis.

S/ N	Variables	Sensitivity variation for the different variables					
	Wind speed (m/s)	5.51	6.64	7.77	8.90	10.03	11.16
	Solar radiation (kWh/m ² /day)	5.84	6.075	6.31	6.545	6.78	7.015
	Minimum SOC of BESS (%)	15	20	25	30	35	40
	Diesel fuel price (\$/L)	0.5	0.675	0.85	1.025	1.20	1.375
	Load demand (kWh/day)	130	133.19	136.38	139.57	142.76	145.95

Table 5

Cases used for simulation.

Case	System type	System configuration
1	Renewable energy-based	PV, WT, BESS, and converter
2	HRES	PV, diesel generator, BESS, and converter
3	HRES	PV, WT, Diesel generator, BESS, and converter

generation, and renewable energy fraction. Emphasizing renewable penetration, the study primarily focused on the proportion of renewable sources integrated into the systems. Additionally, the optimization process determined the optimal quantity of batteries necessary to fulfill power requirements, with the autonomy period representing the maximum duration of sustained load demand. The overall system performance was assessed through critical factors, including surplus power, renewable fraction, unmet load, and the comprehensive system evaluation, guiding the system ranking process.

Three feasible system configurations, classified based on economic perspective, electrical and emission parameters such as CO₂, carbon monoxide (CO), unburned hydrocarbon (UBHC), particular matter (PM), sulfur dioxide (SO₂), and nitrogen oxides (NO₂), along with other configurations, are summarized in Table 6. Each system configuration is evaluated and ranked. The categorized cases are elaborated upon, showcasing several components with varying capacities through a graphical representation. The cost analyses for the various categorized cases are depicted in Fig. 9, illustrating the optimal dispatch for three different HRES configurations. Detailed information regarding the interpretation of electrical RF and excess electricity (EE), along with the specification of the LCOE, is provided in Fig. 10 (a) and (b).

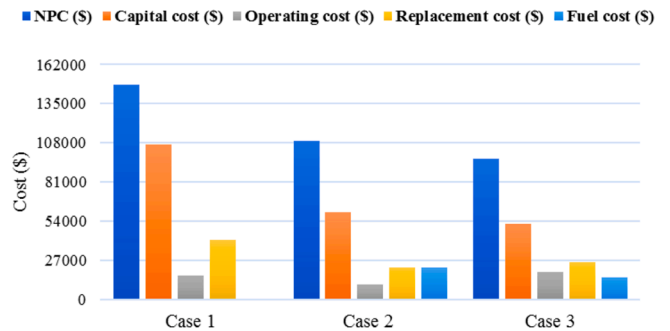
3.1.1. PV/Wind/BESS hybrid energy system (Case 1)

A HES comprising a PV array with a power output of 25 kWp, two wind turbines of 10 kW each, and 71 battery units placed in parallel strings was detailed in this configuration. Additionally, inverters with a capacity of 25 kW were included in the system. According to Table 6, the LCOE for this system is reported to be \$0.2062 per kWh, and the total NPC and operating expenses are stated as \$148,174.5 and \$16,690.97, respectively. This system's initial capital expenditure is \$107,170.51, with no fuel costs. Despite the higher capital cost of the system, the highest LCOE is offered owing to the increased expenses associated with the WT and PV arrays and batteries. The energy generation from PV and wind turbines is quantified as 47,447 kWh/yr (38.7 %) and 75,082 kWh/yr (61.3 %). The annual CO₂ emission of 1,897.88 kg/yr, as indicated in Table 6, is comparatively lower than the emissions in the other configuration studied. This phenomenon can be attributed to a significant increase in the utilization of clean energy sources (renewable penetration), which now constitutes the entire energy supply. It is worth noting that the system has been demonstrated to be economically feasible, as evidenced by a payback period of 3.59 years and a discount

Table 6

Comparative performance analysis of various optimal system configurations.

S/ N	Specification	Constraints	Case 1	Case 2	Case 3
	PV (kWp)		25	25	15
	Wind turbine (kW)		20	–	10
	Diesel generator (kW)		–	25	25
	Battery (kWh)		142	112	72
	System converter (kW)		25	25	16
	Cost (\$)	LCOE (\$/kWh)	0.2062	0.1520	0.0900
		Net present cost	148,174.50	109,327.00	96,899.16
		Capital cost	107,170.51	60,429.79	52,424.59
		Operating cost	16,690.97	10,667.95	18,900.38
		Replacement	41,153.02	22,295.61	25,886.03
		Fuel cost	0.00	21,957.43	15,337.04
		Diesel (L/year)	0.00	3,039	1,418
	Fuel consumption				
	Battery	Usage nominal capacity (kWh)	114	90	58
	performance	Nominal capacity(kWh)	142	112	72
		Energy in (kWh/year)	11,261	25,350	14,485
		Energy out (kWh/year)	10,153	22,892	13,073
		Expected life (year)	15	13.9	15
	Electrical	Renewable fraction (%)	100	82.4	91.8
		Excess electricity (%)	57.6	2.56	25
		Unmet electric load (%)	0.0814	0	0
		Capacity shortage (%)	0.0999	0	0
	Emissions (kg/year)	CO ₂	1,897.88	7,955	3,712
		CO	0	50.1	23.4
		UBHC	0	2.19	1.02
		PM	0	0.304	0.142
		SO ₂	0	19.5	9.09
		NO ₂	0	47.1	22

**Fig. 9.** Optimal dispatch for three different configurations.

payback of 4.07 per year. Furthermore, the system will generate a total profit of \$404,900 over its 25-year operational lifespan.

Based on the International Energy Agency (IEA) and Asian Development Bank, the Emission Factor (EF) was evaluated to reduce CO₂ emissions in PV projects when the grid EF was considered [40,41]. Eqs. (14) and (15) demonstrate that the PV system produces 47,447 kWh/year of power, hence emitting 1,897.88 kgCO₂eq as a baseline emission (BE) as follows:

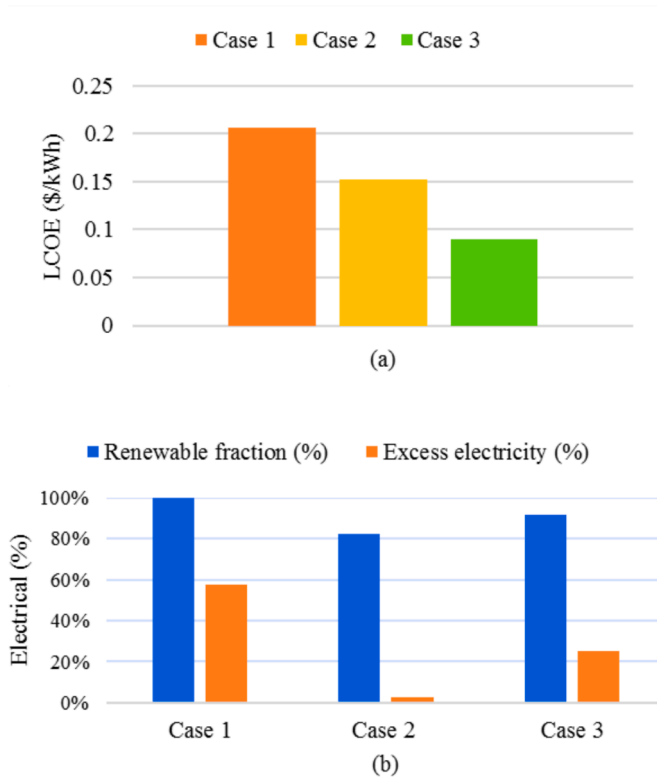


Fig. 10. Comparisons among (a) LCOE and (b) Electrical (RF vs EE) of the different configurations.

$$EE_{grid} = \frac{BEofCO_2(kgortons)}{Annualelectricityproduction\left(\frac{MWh}{Year}\right)} \quad (14)$$

$$EmissionReduction(ER) = BE - PE \quad (15)$$

The BE formula is obtained by multiplying the energy generation from RE projects (REP) with the combined EF of gCO₂eq per MWh or equivalent as follows (see Eq. (16) [42]:

$$BE = energyoutput(REP) * EF \quad (16)$$

The life-cycle carbon footprint of a PV system was approximately 40 gCO₂eq/kWh [43]. This outcome is summarized as follows:

1. Around 60 to 70 % of the carbon result was attributable to the upstream plant processes, which included modules, inverters, supports, and other components.
2. Around 21 to 26 % was contributed by operational activities.
3. Around 5 to 20 % was associated with the disposal or downstream processes of panels.

The carbon footprint of the project or project emission (PE) was approximately 1,897.88 kg CO₂eq emissions annually for 47,447 kWh/year of energy output from a PV system.

3.1.2. PV/DG/BESS hybrid energy system (Case 2)

Referring to Table 6, a 25 kWp PV array, 25 kW DG, and 112 units of storage batteries composed of 56 parallel strings constitute the Hybrid Energy System (HES). Furthermore, the system possesses a 25-kW inverter. The previous configuration's (Case 1) LCOE was substantially lower at \$0.1520/kWh compared to Case 2, which has an LCOE of \$0.2062/kWh. Similarly, the NPC revealed a lower value (26.22 %) than the previous configuration. The reason for this disparity is the slightly lower initial capital cost of 43.61 % compared to Case 1. However, this

hybrid system produced a higher environmental footprint (Case 2) as 7,955 kg/year of CO₂ was emitted, with an RF of 82.4 %. Nevertheless, a mere 8,740 kWh/yr of electricity is generated by the entire power generation from a DG system, reflecting a modest 15.6 % of the total energy generated. On the other hand, 47,447 kWh/yr of energy output is attributed to PV energy production, constituting 84.4 % of the total energy output, as shown in Fig. 11. This configuration witnessed 2.56 % excess electricity production. In contrast, Case 1 observed 57.6 % excess electricity production. Although this system offers less renewable energy penetration with higher pollutant emissions, this configuration is considered inappropriate compared to Case 1.

3.1.3. PV/Wind/DG/BESS hybrid energy system (Case 3)

In configuration 3, a 15 kWp PV array, 10 kW wind turbine, and 25 kW DG are comprised by the HES, serving as a backup power source. Additionally, 72 storage battery units comprise 36 parallel strings and a 16 kW bidirectional inverter. The battery bank's primary role is to satisfy the peak load demands. The LCOE for this system is \$0.0900/kWh, which is slightly lower than that of the other cases under study, bearing in mind that the current LCOE for Somalia's residential consumers is in the range of (0.46–0.81) per kWh. Similarly, the NPC and operating costs are marginally reduced at \$96,899 and \$18,900.38, respectively. This reduction is mainly attributed to the lower initial capital cost of \$52,424.59 compared to other configurations. Nevertheless, the contribution of DG to the total power generation is relatively modest, accounting for only 4,078 kWh (5.82 %) of the total energy produced.

In contrast, the generation of energy from PV and wind turbines is accounted for as 28,468 kWh/yr (40.6 %) and 37,541 kWh/yr (53.6 %), respectively, as depicted in Fig. 11. Notably, the combined power generation from PV, wind, and DG comprises a total power generation of 70,087 kWh (100 %). The charging of surplus electrical energy and its storage in the battery bank is designed for by the RES (PV and WT). Approximately 25 % of the total electrical energy generated is reserved in the battery bank. On the other hand, DG is tailored to be operated and power generated precisely to meet the primary load demand. A battery bank with 72 kWh nominal capacity produced an operational autonomy of 10.1 h in this study. A significantly reduced environmental footprint is exhibited by this HES, with only 3,712 kg/year of CO₂ emitted, which is 53.34 % lower than Case 2, showing an RF of 91.8 %. This substantial reduction was primarily due to the decreased consumption of diesel fuel, which was 83.77 % lower than in Case 2. Despite its slightly higher initial capital cost, as indicated in Table 6, a higher share of RE contribution is boosted by this configuration, making it the most optimal system across all evaluated factors compared to alternative cases.

3.1.4. Technical assessment of optimal configuration

The technical assessment of the optimal configuration involves a detailed examination of both techno-economic and environmental characteristics. The microgrid requires 136 kWh/day, with a peak capacity of 22 kW. The proposed system utilizes various generation sources to supply the electrical load, as depicted in Fig. 12, with a

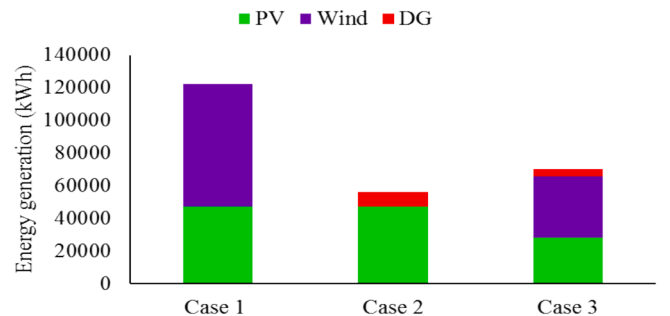


Fig. 11. Contribution of energy generation from different element sources.

breakdown of power contributions from each component. The wind power system emerges as the primary electricity generator, harnessing approximately 53.6 % of the total output due to favorable wind conditions. Following closely, the PV system contributes about 40.6 % of the total electricity, while the DG contributes 5.8 % of the total energy output. Therefore, investors gain insights into the performance, feasibility, and sustainability of the proposed HRES. Equipped with this data-driven analysis, decision-makers can make informed choices regarding investments, resource allocation, and strategic planning, thereby ensuring optimal outcomes aligned with their objectives and priorities.

3.2. Economic assessment

The minimization of costs highlights the significance of economic assessment in the HOMER simulation tool. Commonly utilized economic metrics, such as NPC and LCOE, are utilized in analyzing various system architectures to determine the most cost-effective solution. The NPC, comprising capital costs, replacement costs, O&M costs, fuel costs, and salvage costs, is commonly used. The total NPC for each system case is depicted in Fig. 9. Case 1 had the largest NPC amount among the three system cases, with Case 2 ranked second in NPC. The higher cost of the former was primarily attributed to its substantial initial capital investment, with the most significant fraction being allocated to PV systems, WT, and BESS.

Conversely, the second type incurred a higher cost due to the O&M costs of the WT. In Case 2, a higher share of the NPC was attributed to the initial capital cost. No expenses were incurred for O&M due to the absence of the WT in this configuration. A lower replacement cost was observed because no required replacement costs were associated with the PV and DG systems. The cost trend in Case 3 presents a different pattern from the cost results obtained in Cases 1 and 2. In Case 3, fuel and operating costs were increased, although there was a lower initial capital cost compared to the other analyzed configurations.

Moreover, in Case 3, an NPC of \$96,899.16 and operating costs of \$18,900.38 were determined. When Case 3 is analyzed, it becomes evident that the most significant portion of the operating costs was attributed to O&M expenditures, specifically from the WT, followed by DG. At the same time, the converter accounted for the most minuscule fraction of the overall NPC. Regarding Case 3's NPC, the most substantial contributor to the total NPC is the capital cost, closely followed by the replacement costs. This is primarily because of the high initial capital investments coupled with PV, WT, DG, and BESS. Subsequently, a significant role was played by O&M costs, with the largest share being

shouldered by WT. Conversely, expenditures related to running costs and fuel usage remained relatively small. Notably, the NPC in the Case 3 system is the lowest among the configurations studied.

From an economic perspective, the system boasts an impressive return on investment (ROI) of 88.6 % and an internal rate of return (IRR) of 88.8 %. The ROI and IRR values strongly indicate that the system is a substantially profitable investment. Conversely, incorporating a WT to increase system feasibility can reduce roughly 11.37 % and 40.79 % in NPC and LCOE, respectively, compared to case 2. Furthermore, a 53.34 % reduction in fuel consumption can be achieved by adding a backup (DG) (see Table 6). The backup DG system can also be deployed to fulfill the energy requirements when RER (PV and WT) are inaccessible. In addition, surplus power generated by RE sources can be used to charge the BESS, enabling it to serve as a backup in conjunction with DG or WT to meet fluctuating load demands.

3.3. Environmental impact assessment

The economic viability of hybrid systems that integrate RE and non-renewable energy (non-RE) sources is deemed crucial, and the evaluation of such viability is contingent upon considering the quantity of greenhouse gases (GHG) and pollutant emissions. The CO₂ and other pollutants released during power conversion from fuel combustion, encompassing carbon dioxide, nitrogen oxide, carbon monoxide, unburned hydrocarbons, and sulfur dioxide, were considered in this study. The hazardous emission assessment included the consideration of diesel fuel, owing to a direct correlation between total emissions and global warming. The relationship between the release of emissions and the process of diesel fuel burning was examined in this study. Two primary components are necessitated for the computation of emissions: the yearly consumption of diesel and emission factors. These factors are determined by parameters such as density, carbon and sulfur content, and low heating value. The pollutant emissions of several analyzed system cases are presented in Table 6. The calculation of total emissions often involves the utilization of emission values for several components, such as DG and RE sources.

In most cases, consideration is given to environmental impacts to ensure the sustainability of the system configuration. A total pollutant emission of 3,712 kg/yr was produced by the optimized hybrid PV/WT/DG/BESS (Case 3) system configuration. In contrast, 1,897.88 kg/yr (Case 1) and 7,955 kg/yr (Case 2) were generated by the PV/WT/BESS and PV/DG/BESS system configurations, respectively. The PV/WT/DG/BESS (Case 3) model exhibited the lowest emissions across all emission

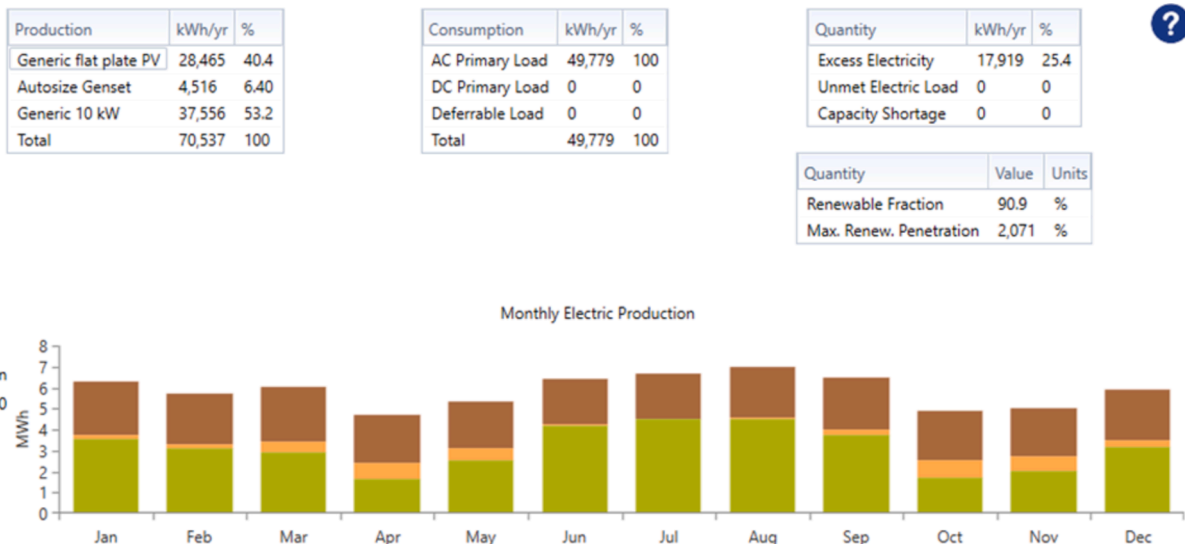


Fig. 12. Optimal monthly electricity production and power consumption.

components. This configuration achieved an RF of 91.8 % because the system primarily relied on RE sources. A negligible consumption of fossil fuels is resulted from the presence of DG. This confirms the lower fuel consumption compared to the other configurations studied (Case 2).

In addition, the DG system is integrated in Case 2, causing a reduction in its renewable share by approximately 21.36 % and increasing its annual fuel consumption from 0 to 3039 L. According to the fuel consumption rate in system Case 2, the PV/DG/BESS system is operated for 699 h/year, with an average daily consumption of 8.33 L and 0.347 L/hour to meet the system load demands. This leads to yearly CO₂ emissions of approximately 7,955 kg. The DG for system Case 3 is operated for 326 h/year and consumes 1,418 L/year, resulting in an average daily consumption of 3.88 L and an hourly consumption of 0.162 L of diesel fuel. Consequently, the lowest yearly CO₂ emissions, amounting to 3,712 kg/yr, are produced by system Case 3 compared to the other system cases. Incorporating wind power into system Case 3 leads to an approximately 53.34 % reduction in annual fuel consumption and CO₂ emissions. In Case 1, the system demonstrates the utilization of 100 % PV production, leading to a yearly energy output of 47,447 kWh. According to the reference [42], the lifecycle carbon footprint of the RE system is estimated to be around 40 g CO₂eq/kWh. The annual carbon footprint of the project, or Project Emission (PE), was assessed to be roughly 1,897.88 kg CO₂eq. This calculation was based on the energy output of 47,447 kWh/yr produced by the PV source. An environmental impact concerning the emissions of various systems (Case 3) was exhibited by configuration Case 1. While an inclusive approach regarding RER was adopted by this system configuration (Case 1), the level of competence observed in other system cases concerning techno-economic factors has unfortunately not been achieved. A satisfactory rate of CO₂ emissions was demonstrated by the optimal configuration comprising PV/WT/DG/BESS. The CO₂ emissions of system Case 3 were successfully reduced by approximately 53.34 % compared to configuration Case 2, indicating a significant improvement in environmentally friendly.

3.4. Sensitivity analysis

The impact of varying independent variables on the optimal system's O&M costs, fuel consumption, renewables, and load demand was evaluated through a sensitivity analysis to determine the strength and reliability of the study. The sensitivity factors selected for this analysis were considered, including wind speed, global solar irradiation, minimum state of charge (BESS SOC_{min}), diesel price, and community load demand. For each sensitivity variable, including the load demand growth,

the base case value was either decreased or increased. The initial values for wind speed, global solar irradiation, minimum state of charge (BESS SOC_{min}), diesel price, and community load demand were 7.77 m/s, 6.31 kWh/m²/day, 20 %, \$0.85/L, and 136.38 kWh/day, respectively. Consequently, assessing how these modifications might affect the system's efficiency and specifications is deemed necessary. This aids in evaluating the viability and dependability of the project. The optimal system operating, fuel, and maintenance costs are considered in the sensitivity analysis, as depicted in Table 6.

The sensitivity analysis reveals the substantial influence of the increase in load demand on system cost operation, fuel, and maintenance. The impact of fluctuating fuel prices on system costs, renewable penetration, and fuel is depicted in Fig. 13. Additionally, changes in gasoline prices significantly affect fuel and operating costs but have a minor impact on O&M costs. An increase in fuel prices from \$0.5 to 1.357 per liter increased operational and fuel expenses by approximately 5.59 % and 17.07 %, respectively, while the O&M costs increased by 8.01 %. This is attributed to the cost of running the DG when diesel fuel prices rise.

The LF dispatch control approach was employed in this study to minimize operational expenses, allowing the battery to be charged using RE sources (PV and wind power). The battery's SOC_{min} was set between 20 % and 45 %, contributing to the overall lifespan of the battery. This was accompanied by a reduction in O&M expenditures for the DG from \$244 per year to \$264 per year. It was observed that greater reliance on RE to meet the load demand, resulting in reduced fuel consumption and decreased DG-related costs, was achieved by increasing the battery's SOC_{min}. The yearly fuel expenditure was also reduced from \$1,120 to \$709. Typically, a range of 30 % to 50 % SOC is chosen to prevent excessive discharge, which could potentially cause damage to the battery.

Furthermore, in all examined sensitivity possibilities, it is demonstrated by the study that the NPC of the optimal system configuration is increased as both the diesel fuel price and the load consumption rise (see Fig. 14). For example, the NPC is increased by approximately 6.36 % as fuel prices increase from (\$0.5 to \$1.375)/L and by roughly 57.97 % due to a load demand growth of around 12.27 kWh/day compared to the current load consumption. When comparing them, it is observed that the LCOE value increases linearly with rising fuel prices but reduces as load demand rises. Additionally, a fuel consumption rate of 1,418 L/yr is noted for the most efficient system, significantly lower than the rates for other configurations (PV-DG-BESS) explored, which exceeded 53.34 L/yr. There is also a noticeable correlation between the increase in load consumption and the rational increase in fuel prices. This correlation is

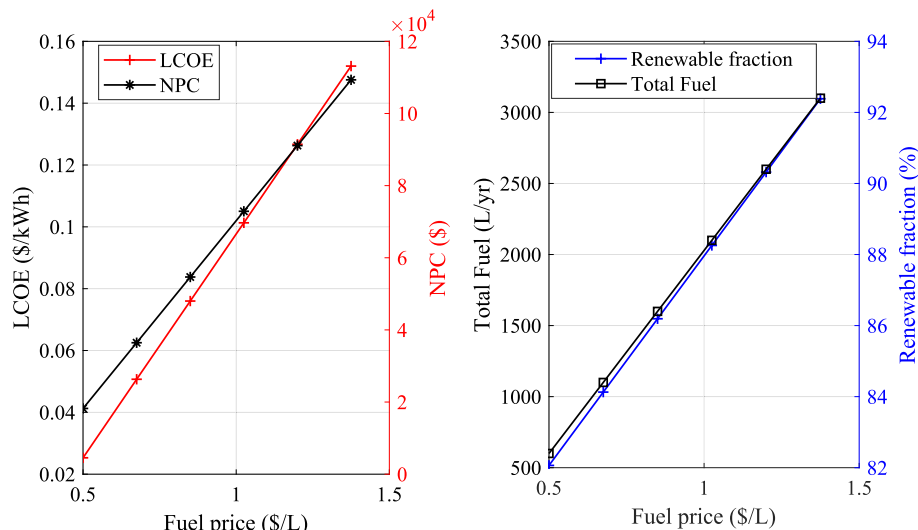


Fig. 13. The impact of fluctuating fuel prices on system costs, renewable penetration, and fuel.

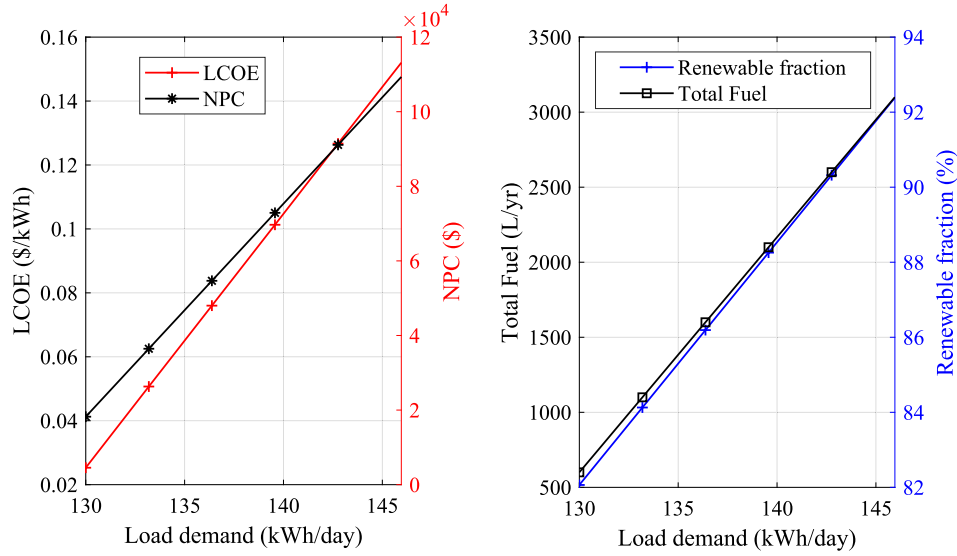


Fig. 14. The impact of load demand growth on system costs, renewable penetration, and fuel.

particularly evident when the fluctuations in fuel prices and the community load requirements are considered, as illustrated in Fig. 14. For instance, when the load demand increased from 130 kWh/day to 145.96 kWh/day, the total fuel consumption increased by about 3096 L/yr from 1418 L/yr. Simultaneously, the RE penetration was decreased by approximately 14.56 % from the primary penetration rate of 91.8 %.

The impact of different solar radiation levels on system costs, the RF used, and the overall fuel consumption is illustrated in Fig. 15, and a reduction of approximately 15.02 %, 36.67 %, and 63.61 % in NPC, LCOE, and total fuel utilized, respectively, is revealed by the statistics. An increase of 7.19 % in the share of the RF was observed as solar radiation at the selected location increased from 5.84 to 7.015 kWh/m²/day. The decrease in cost parameters can be attributed to reduced fuel consumption, leading to reduced diesel expenses.

The influence of varying wind speed on system costs, the RF consumed, and the total fuel consumed is depicted in Fig. 16. A reduction of approximately 15.02 % in the NPC, 36.67 % in the LCOE, and 11.53 % in the total fuel consumption is revealed by the data. As wind speed at the selected site is increased by 5.51 to 11.51 m/s, a 19.89 % increase in the penetration of the RF is observed. The decrease in cost

constraints can be attributed to the reduction in overall fuel consumption and the consequent decline in diesel-related costs.

In this study, we investigated the effects of varying the minimum state of charge SOC_{min} of a battery within the range of 15 % to 45 % on the operational performance of the optimal design. The consequences of this variation were analyzed with a focus on critical metrics, including system costs, annual fuel consumption, minimum state of charge of the battery, and renewable penetration, as depicted in Fig. 17. The study's findings revealed notable trends. As the SOC_{min} of the battery was increased from 15 % to 45 %, there was a corresponding increase in the values of NPC, LCOE, and annual total fuel consumption, showing gains of approximately 4.52 %, 5.56 %, and 13.03 %, respectively. Concurrently, an evident reduction in the RF by about 5.88 % was observed with the elevation of the battery state of charge. This implies that higher SOC_{min} levels decrease the fraction of energy taken from renewable sources within the overall system.

A vital inference drawn from these results is the potential environmental impact. The increase in annual fuel consumption, accompanied by the decrease in RF, suggests an increase in polluting emissions. Consequently, this poses a significant environmental challenge,

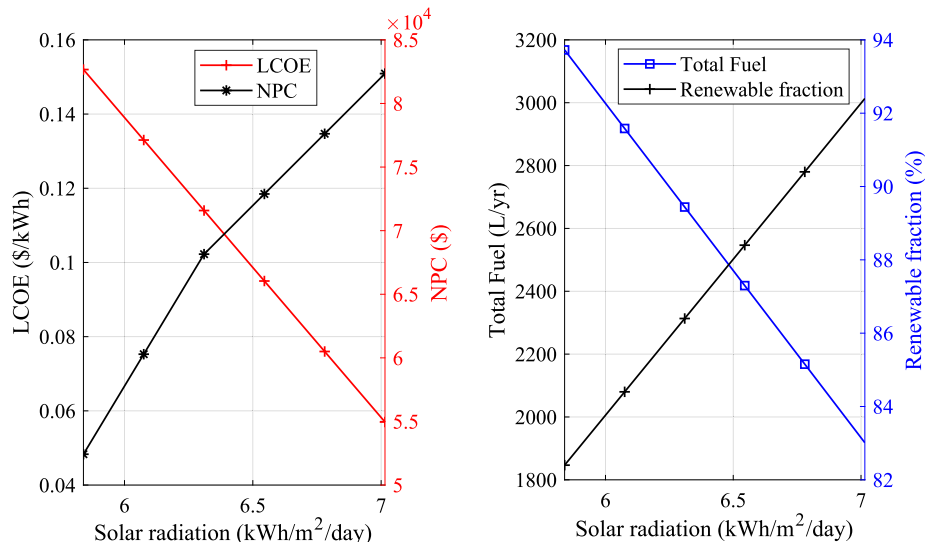


Fig. 15. The effect of fluctuating solar radiation on system costs, renewable penetration, and fuel.

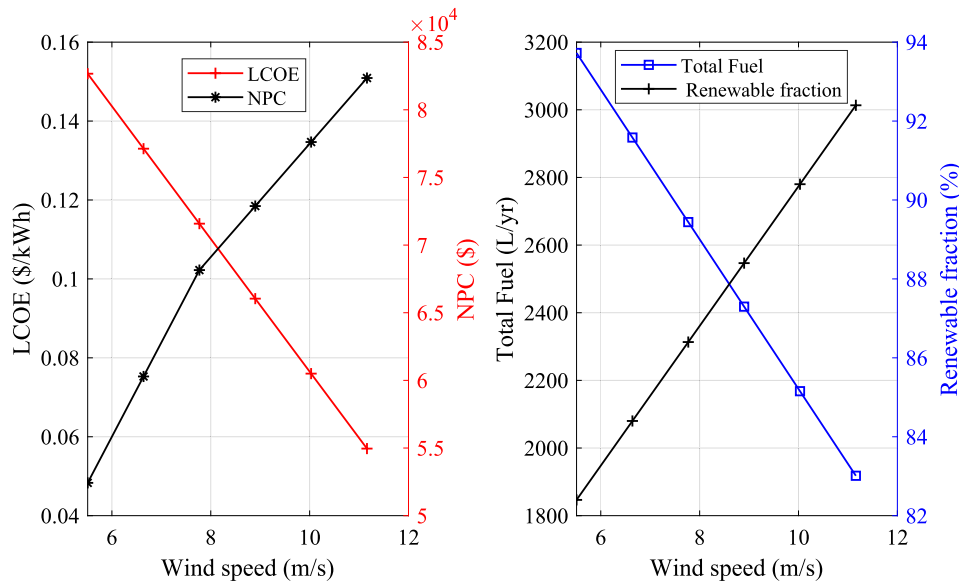


Fig. 16. The impact of fluctuating wind speed on the system costs, RE penetration, and fuel usage.

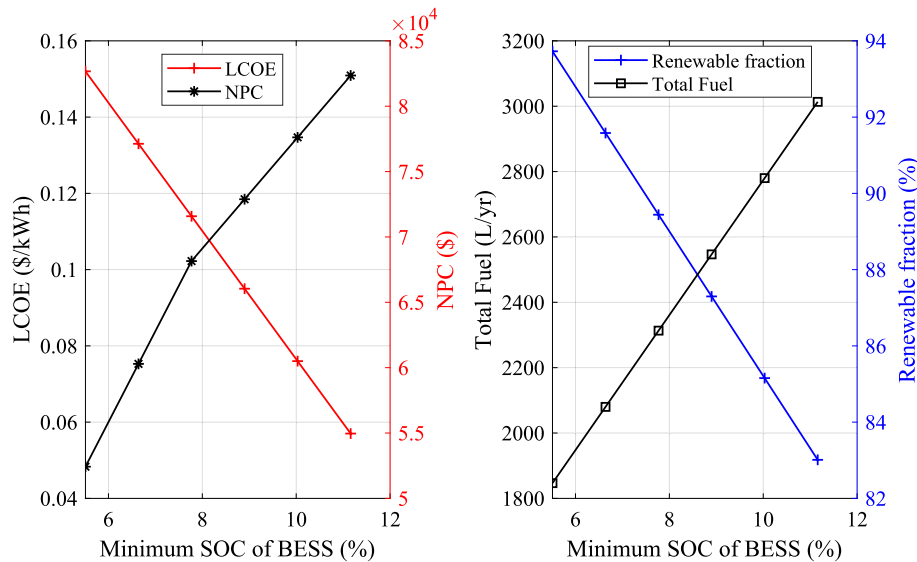


Fig. 17. The impact of fluctuating the BESS SOC_{min} on the system costs, RE penetration, and fuel usage.

emphasizing the need for a careful balance between battery state of charge and sustainable energy practices in the overall system design.

3.5. Comparison of proposed HRES with existing systems

Current research underscores the imperative for researchers to explore various locations and assess location-specific RE potential and meteorological conditions, which are crucial for evaluating the system's viability. Consequently, selecting a methodology that significantly contributes to obtaining accurate findings is essential. Scholars predominantly focus on implementing off-grid HRES, which is universally prevalent in municipalities [44–47]. Recent global studies have revealed a notable gap, with no research conducted on HRES in Somalia [48]. Various studies utilizing HRES have been performed in different locations, revealing that integrating PV/WT/DG with the storage system is more economical than the baseline system. M. Hossain *et al.* evaluated an HRES for a resort in Malaysia's South China Sea region, focusing on technical and economic aspects. The study achieved an LCOE of \$0.279/

kWh, RF of 41.6 %, and annual CO₂ emissions totaling 2,571,131 kg/year. This was accomplished by implementing a proposed PV/WT/DG/BESS system with converters [36]. Another study conducted by M. F. Ishraque *et al.* reveals that the LF dispatch strategy excels in both economic and environmental aspects, exhibiting the lowest LCOE at \$0.219/kWh, NPC of \$160,360, and CO₂ emissions of 3,297 kg/year. The proposed hybrid microgrid systems demonstrate significant improvements, including a 42.37 % reduction in COE, a 46 % decrease in NPC, and an impressive 99.99 % reduction in CO₂ emissions compared to conventional fossil fuel-based bases [49]. A comparison between the proposed HRES and existing alternatives shows significant differences between the LCOE and RF. Additionally, the researcher appears to have overlooked the critical aspect of addressing the environmental impact of standalone RE systems, especially PV systems.

4. Conclusions

This paper aims to unravel the techno-economic perspective of HRES

in providing sustainable power to rural communities in sub-Saharan Africa, with a focus on Somalia, where communities are far from the electrical grid and experience energy shortages. A comprehensive assessment was undertaken to analyze various system scenarios' performance and operational strategies. When evaluating the feasibility of financing energy infrastructure, stakeholders carefully consider several crucial characteristics, including the NPC, LCOE, payback period, return on investment, internal rate of return, and environmental challenges. The HRES model was validated using HOMER software. To ensure the reliability of the results, researchers conducted comprehensive techno-economic analyses, including a sensitivity study and an assessment of system performance.

The simulation results indicate that the PV/WT/DG/BESS model, comprising a 15 kWp PV array, 10 kW WT, 25 kW DG, 72 units of storage batteries, and a 16-kW bidirectional inverter, is optimized with an NPC and LCOE of \$96,899.16 and \$0.090/kWh, respectively. The NPC shows a significant decrease of 34.60 % and 11.37 % compared to the NPC for system Cases 1 and 2. Due to the substantial share of the RF in this configuration (91.8 %), the system utilizes only 1,418 L of diesel. Therefore, the proposed system is the most efficient power supply choice for achieving sustainable rural electrification in Somalia, emitting a cumulative total of 3,712 kg/year in pollutant emissions. Moreover, we evaluate the optimum hybrid configuration through sensitivity analysis using parameters such as NPC, LCOE, RF, fuel consumption, and BESS SOC_{min}. We observe that variations in these parameters at different rates impact the NPC. Substantially, the findings of this work can be interpreted as a primary step towards assessing more sectors as part of the path towards increasing the share of RE systems in Somalia.

CRediT authorship contribution statement

Abdullahi Mohamed Samatar: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Project administration. **Saad Mekhilef:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization. **Hazlie Mokhlis:** Writing – review & editing, Validation, Supervision. **Mostefa Kermadi:** Writing – review & editing, Formal analysis. **Obaid Alshammari:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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